

## MILFORD HAVEN PORT AUTHORITY, United Kingdom

WMO: 036040

Lat: 51.700N

Lon: 5.050W

Elev: 32

StdP: 100.94

Time Zone: 0.00 (EUW)

Period: 94-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |     | Humidification DP/MCDB and HR |     |      |      |     |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|-----|-------------------------------|-----|------|------|-----|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |     | 99.6%                         |     |      | 99%  |     |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99% | DP                            | HR  | MCDB | DP   | HR  | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c) | (d)                           | (e) | (f)  | (g)  | (h) | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 2             | -1.1       | 0.3 | -5.2                          | 2.4 | 0.6  | -3.6 | 2.8 | 1.7  | 17.0                  | 7.8  | 15.3 | 8.1  | 3.8                   | 50   | 0.661 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 8             | 5.4                    | 22.5            | 17.5 | 20.8 | 16.7 | 19.5 | 16.1 | 18.3                | 21.1 | 17.5 | 19.6 | 16.8 | 18.5 | 4.0                  | 120  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| 17.3                            | 12.4 | 18.8 | 16.7 | 11.9 | 18.1 | 16.1 | 11.5 | 17.6 | 51.6          | 21.2 | 49.2 | 19.7 | 47.4 | 18.6 | 22.8              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |      |     |
|-------------------|------|------|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|------|-----|
|                   |      |      | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |      |     |
| 1%                | 2.5% | 5%   | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  |      |     |
| (a)               | (b)  | (c)  | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |      |     |
| (4) 14.0          | 12.4 | 11.0 | DB                         | -2.6 | 25.5               | 1.5 | 1.8                                                | -3.7 | 26.8       | -4.6 | 27.9       | -5.5 | 28.9       | -6.6 | 30.2 | (4) |
| (5)               |      |      | WB                         | -3.6 | 20.0               | 1.5 | 1.3                                                | -4.7 | 20.9       | -5.6 | 21.7       | -6.5 | 22.4       | -7.6 | 23.4 | (5) |

## Monthly Climatic Design Conditions

|                                                                                  |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec  |
|----------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|                                                                                  |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)  |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 11.0   | 6.6   | 6.5   | 7.4   | 9.2   | 11.8  | 14.2  | 16.1  | 16.1  | 14.5  | 12.2  | 9.4   | 7.3  |
|                                                                                  | DBStd       | 4.19   | 2.77  | 2.40  | 2.21  | 2.18  | 2.21  | 1.98  | 1.87  | 1.58  | 1.66  | 2.16  | 2.52  | 2.99 |
|                                                                                  | HDD10.0     | 469    | 107   | 99    | 83    | 38    | 7     | 0     | 0     | 0     | 0     | 7     | 40    | 89   |
|                                                                                  | HDD18.3     | 2694   | 365   | 332   | 339   | 273   | 202   | 124   | 74    | 71    | 114   | 189   | 269   | 342  |
|                                                                                  | CDD10.0     | 826    | 1     | 1     | 2     | 14    | 63    | 127   | 189   | 191   | 136   | 76    | 21    | 6    |
|                                                                                  | CDD18.3     | 9      | 0     | 0     | 0     | 0     | 0     | 2     | 5     | 3     | 0     | 0     | 0     | 0    |
|                                                                                  | CDH23.3     | 33     | 0     | 0     | 0     | 0     | 0     | 6     | 17    | 11    | 0     | 0     | 0     | 0    |
|                                                                                  | CDH26.7     | 2      | 0     | 0     | 0     | 0     | 0     | 0     | 1     | 1     | 0     | 0     | 0     | 0    |
| Wind                                                                             | WSAvg       | 5.4    | 6.3   | 6.1   | 5.5   | 5.0   | 4.8   | 4.5   | 4.3   | 4.6   | 5.0   | 5.9   | 6.0   | 6.3  |
| Precipitation                                                                    | PrecAvg     | 861    | 89    | 65    | 66    | 54    | 48    | 52    | 49    | 73    | 74    | 93    | 102   | 97   |
|                                                                                  | PrecMax     | 1123   | 182   | 155   | 158   | 138   | 120   | 115   | 136   | 181   | 147   | 208   | 257   | 179  |
|                                                                                  | PrecMin     | 618    | 13    | 2     | 10    | 10    | 5     | 6     | 3     | 0     | 11    | 19    | 30    | 27   |
|                                                                                  | PrecStd     | 118    | 40    | 36    | 31    | 30    | 27    | 31    | 27    | 39    | 38    | 48    | 48    | 41   |
| Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 11.7  | 11.4  | 14.3  | 18.3  | 21.5  | 23.9  | 25.4  | 24.7  | 21.2  | 17.7  | 14.6  | 13.0 |
|                                                                                  |             | MCWB   | 10.7  | 9.6   | 10.2  | 13.0  | 15.9  | 17.4  | 18.4  | 19.1  | 17.4  | 15.1  | 13.5  | 12.1 |
|                                                                                  | 2%          | DB     | 11.1  | 10.6  | 12.1  | 15.7  | 19.1  | 21.2  | 22.9  | 21.5  | 19.2  | 16.5  | 13.9  | 12.4 |
|                                                                                  |             | MCWB   | 10.3  | 9.5   | 9.4   | 11.7  | 14.5  | 16.3  | 17.6  | 17.6  | 16.1  | 14.7  | 13.0  | 11.5 |
|                                                                                  | 5%          | DB     | 10.6  | 10.1  | 11.0  | 13.8  | 17.1  | 19.2  | 20.9  | 20.0  | 18.2  | 15.7  | 13.3  | 11.8 |
|                                                                                  |             | MCWB   | 9.8   | 9.2   | 9.0   | 10.5  | 13.5  | 15.3  | 16.9  | 16.6  | 15.5  | 14.3  | 12.4  | 11.0 |
|                                                                                  | 10%         | DB     | 10.1  | 9.6   | 10.2  | 12.5  | 15.4  | 17.8  | 19.4  | 18.8  | 17.3  | 15.1  | 12.8  | 11.2 |
|                                                                                  |             | MCWB   | 9.3   | 8.7   | 8.8   | 9.9   | 12.5  | 14.6  | 16.2  | 16.1  | 15.1  | 13.9  | 11.8  | 10.3 |
| Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 11.2  | 10.6  | 11.1  | 13.7  | 16.6  | 18.5  | 19.3  | 19.7  | 18.2  | 16.0  | 14.1  | 12.6 |
|                                                                                  |             | MCDB   | 11.3  | 10.8  | 12.4  | 17.1  | 20.3  | 22.5  | 23.6  | 23.4  | 19.9  | 16.9  | 14.2  | 12.7 |
|                                                                                  | 2%          | WB     | 10.7  | 10.0  | 10.2  | 12.2  | 15.1  | 16.9  | 18.2  | 18.3  | 17.0  | 15.3  | 13.4  | 12.0 |
|                                                                                  |             | MCDB   | 10.9  | 10.3  | 11.1  | 14.7  | 18.3  | 20.2  | 21.6  | 20.5  | 18.2  | 16.0  | 13.6  | 12.3 |
|                                                                                  | 5%          | WB     | 10.1  | 9.5   | 9.7   | 11.2  | 14.0  | 15.9  | 17.4  | 17.5  | 16.4  | 14.8  | 12.8  | 11.3 |
|                                                                                  |             | MCDB   | 10.4  | 9.9   | 10.5  | 13.0  | 16.2  | 18.3  | 20.0  | 18.9  | 17.4  | 15.4  | 13.2  | 11.6 |
|                                                                                  | 10%         | WB     | 9.6   | 9.0   | 9.2   | 10.4  | 13.1  | 15.1  | 16.7  | 16.8  | 15.8  | 14.2  | 12.1  | 10.6 |
|                                                                                  |             | MCDB   | 10.0  | 9.5   | 10.0  | 11.9  | 14.9  | 17.1  | 18.7  | 18.1  | 16.8  | 15.0  | 12.7  | 11.1 |
| Mean Daily<br>Temperature<br>Range                                               | 5% DB       | MDBR   | 3.7   | 4.0   | 4.8   | 5.7   | 6.1   | 5.8   | 5.8   | 5.4   | 5.3   | 4.4   | 3.9   | 3.7  |
|                                                                                  |             | MCDBR  | 3.4   | 3.8   | 5.7   | 7.8   | 8.3   | 8.3   | 7.1   | 6.4   | 4.5   | 3.4   | 3.3   |      |
|                                                                                  |             | MCWBR  | 3.8   | 3.4   | 4.1   | 4.9   | 5.1   | 5.0   | 4.6   | 4.1   | 4.2   | 3.5   | 3.5   | 3.9  |
|                                                                                  | 5% WB       | MCDBR  | 3.4   | 3.2   | 4.5   | 6.5   | 7.2   | 7.2   | 7.1   | 5.9   | 5.2   | 3.7   | 3.2   | 3.3  |
|                                                                                  |             | MCWBR  | 4.1   | 3.4   | 3.8   | 4.4   | 4.7   | 4.6   | 4.3   | 4.0   | 4.1   | 3.4   | 3.6   | 4.1  |
| Clear-Sky<br>Solar<br>Irradiance                                                 | taub        | 0.336  | 0.347 | 0.375 | 0.386 | 0.395 | 0.381 | 0.387 | 0.370 | 0.365 | 0.356 | 0.334 | 0.325 |      |
|                                                                                  | taud        | 2.329  | 2.379 | 2.328 | 2.330 | 2.352 | 2.422 | 2.410 | 2.473 | 2.474 | 2.444 | 2.385 | 2.306 |      |
|                                                                                  | Ebn at Noon | 657    | 769   | 814   | 851   | 859   | 874   | 862   | 860   | 822   | 749   | 658   | 598   |      |
|                                                                                  | Edh at Noon | 65     | 81    | 102   | 114   | 117   | 110   | 109   | 98    | 88    | 75    | 61    | 57    |      |
| All-Sky Solar<br>Radiation                                                       | RadAvg      | 0.74   | 1.51  | 2.75  | 4.40  | 5.40  | 5.88  | 5.36  | 4.47  | 3.31  | 1.83  | 0.92  | 0.56  |      |
|                                                                                  | RadStd      | 0.07   | 0.14  | 0.24  | 0.31  | 0.34  | 0.49  | 0.44  | 0.30  | 0.24  | 0.15  | 0.07  | 0.07  |      |

## Historical Trends

|                              |  | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|------------------------------|--|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|                              |  |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
|                              |  | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)     | (l)     | (m)     |
| (46) Station Only            |  | +0.27 | +1.10   | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | +56     | N/A     |
| (47) Regional (41 neighbors) |  | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |

Nomenclature: See separate page